

Debajyoti Das

+917001921285 | debajyoti02032000@gmail.com | /in/debajyoti-das | portfolio

PROFESSIONAL SUMMARY

Software Engineer (Agentic AI Engineer) specializing in Generative AI, Agentic Systems, and intelligent enterprise applications. Over the last 2+ years, I have led the development of production-scale AI solutions across pharmaceutical and analytics domains, transforming manual business processes into autonomous AI-driven workflows. Skilled at taking products from ideation to production using modern LLM architectures, RAG pipelines, and cloud-native systems, with demonstrated impact in scalability, operational efficiency, and business adoption.

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL

Libraries & Frameworks: Scikit-learn, TensorFlow, PyTorch, FastAPI, LangChain, LangGraph, CrewAI, OpenAI sdk Agents, AutoGen, Pydantic AI, Agno, Semantic Kernel

Cloud & DevOps: AWS (Lambda, Bedrock, S3, SQS, EC2, ECR, RDS, OpenSearch, API Gateway, Secrets Manager, Parameter Store, K8s, Cloudwatch, ALB), Docker

Developer Tools: Claude code, Git, GitHub, VS Code, Jupyter Notebook, Power BI Desktop (Basic)

Focus Areas: Generative AI, Machine Learning, NLP, Data Science, Backend Engineering, MLOps

Soft Skills: Problem Solving, Leadership, Communication, Adaptability, Time Management

WORK EXPERIENCE

UsefulBI Corporation | Software Engineer II Feb 2024 – Present
Bangalore, India (Onsite)

AgentVerse – Multi-Agent Pharmaceutical Intelligence Platform Mar 2026 – Present

- Developed and productionized a multi-agent GenAI platform for domain-specific pharmaceutical assistance across HIV, Pharmacovigilance, Medical Literature, and Clinical Guidelines using Python-based orchestration.
- Designed and implemented a custom RAG architecture using Claude Sonnet 4.5, Titan v2 Embeddings, and OpenSearch Hybrid Search, with in-house document chunking, embedding, indexing, and retrieval; integrated AWS Knowledge Bases Web Crawler and S3 for web-content ingestion.
- Built an evaluation and observability pipeline using Amazon RDS and AWS Bedrock Evaluation, capturing agent interactions and production logs to measure retrieval quality, grounding, and response accuracy across agents.

Clinical Trial Document Automation Platform Aug 2024 – Aug 2025

- Built a production GenAI platform for automated generation of ICF and PLPS clinical trial documents, supporting 100+ concurrent users.
- Designed scalable AWS backend using FastAPI, Lambda, SQS, and RDS; implemented RAG with Titan Embeddings, OpenSearch, and Claude Sonnet 3.5/3.7/4.5.
- Developed an LLM-driven TOC Agent and in-context learning workflow, reducing manual medical writing effort by 90% and enabling multi-model inference at scale.

Conversion Rate Optimization Feb 2024 – Aug 2024

- Built a classification model to predict trial users likely to convert to paid subscriptions using behavioral engagement features.
- Enabled targeted reminders, engagement nudges, and offers to improve trial-to-paid conversion.

EDUCATION

National Institute of Technology Kurukshetra, Haryana Oct 2021 – Jul 2024
Master of Computer Applications CGPA: 8.7